

General Features

The central units i101, i102 and i103 are the core of all standard/custom ionia[™] systems. The space-saving form factor allows the design of a compact train controller unit.

Huge processing power for any application is provided by the integrated high performance ARM[®] Cortex[™]-A8 processor (core speed of up to 720 MHz) and 256 MB DDR2-SDRAM. Accessing the train bus (ETH, MVB or CAN) is done directly, without any additional card.

The modules offer a fast and secure point-to-point connection with a maximum of 23 extension modules (input/output or communication). This way, small and large train applications (I/O count) can be addressed easily with one central unit.

A built-in supervisory mechanism called «Condition Monitoring» can detect and record any off-limit usage, such as over-voltage, overcurrent, supply interruption, overtemperature, etc. as well as selected user-defined values.

The service interfaces, through Ethernet and USB, offers quick access to diagnostic information and all recorded data. The module offers a power save mode, used e.g. in a WTB Gateway. Additionally there is an optional display interface available.

Developing the train management application is done easily and time-saving by one of the five methods defined by IEC 61131.

i101, i102 and i103 are designed for harsh rolling stock environments and are fully compliant to EN 50155.

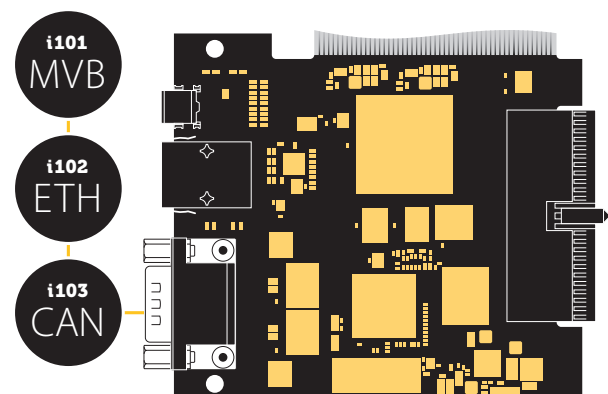
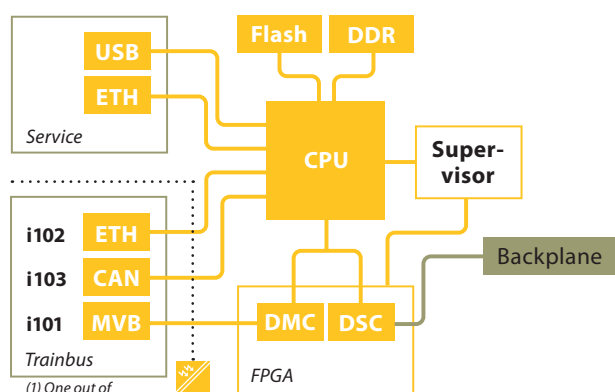
Key Benefits

- Complex applications running on a high-performance 32bit ARM[®] processor
- 100Mbit «Serial Star Controller»: no need for physical module addressing
- Up to 23 extension modules
- IEC61131 for fast and easy application programming
- «Condition Monitoring» for detailed diagnostics through USB and ETH
- One architecture suitable for all networks: Ethernet, MVB and CAN
- Complies to EN 50155, EN 50121, IEC 61375, IEC 61131
- No capacitor replacement necessary

Application examples

- Real-time Vehicle Control Unit
- TCN gateway management
- HVAC controller
- Compact coach controller
- Door controller unit
- Toilet controller unit
- Ambient light controller
- Multiple communication gateways
- Driver machine interface

i101, i102, 103 Hardware Architecture



Life Cycle Cost

The i101, i102 and i103 have been strictly designed without electrolytic capacitors to avoid service cost for changing capacitors. The use of an FPGA with included duagon own soft-IP minimizes the risk and costs in case of component ob-

solescence. Data access and firmware updates are easily done through a service interface or directly via one of the available train buses.

Technical Data

Processor System	– 32 Bit ARM® Cortex™-A8 running at 720 MHz – 256 MB DDR2-RAM – 64 kB non volatile memory for remaining data (up to 1 MB available) – 32 MB boot flash memory – 1 GB SLC flash file system (8 GB available) – RTC with minimum 10 years battery life time	Backplane Communication	– 100 Mbit duagon serial star controller for connecting up to 23 modules – Communication Link with watchdog supervision – Configurable IO scan cycle time minimum: 1 ms
OS	– Realtime Linux	Diagnostics/Service	– Ethernet 10/100 Mbit on RJ45 (only diagnostics and service), DHCP – Webserver for diagnostics and control – USB 2.0 high-speed OTG Type miniAB connector for easy monitoring – Diagnostic LEDs indicating power, system OK, ETH-, USB-, consist network activity
Fieldbus Connections	– MVB SUB-D, EMD or ESD+ up to Class 4 according to IEC61375 – ETH M12 10/100 Mbit/s (Profinet, IPTcom/TRDP, CIP, TCP/IP, UDP), IEEE 802.3 – CAN SUB-D, (CAN, CANopen Master/Slave), EN 50325 – Field bus configuration programmable (IEC 61131) – All interfaces galvanically isolated (ETH 1500 Vac, CAN/MVB 1000 Vac)	Power Consumption	– $P_{max} < 8 \text{ W}$; $P_{save} < 1 \text{ W}$ (in sleep mode) – Power supply via backplane – Interruption Class S2 with ceramic capacitors
Display Interface	– DVI display interface* with touch screen support*	Operating Conditions	– Ambient temperature: –40 to +85 °C (EN 50155, Class TX) – Relative humidity: 75%, max 95% for 30 days per year (conformal coating) EN 60068 – Shock and vibration: According to IEC 61373 category 1, class B – EMI: According to EN 50121 and EN 50155
Programming	– IEC 61131 compliant executor and IDE supporting LD, FBD, ST, IL and SFC – Firmware download through service interface – Firmware upgrade through fieldbus activatable by customer	Dimensions/Weight	– 8TE (i101), 5TE (i102), 4TE (i103) x 92 x 86 mm – 120g
Security / Watchdog	– Off-chip Watchdog – External supervisor for voltage and temp monitoring → «Condition Monitoring»	Environment	– Fully compliant with RoHS and REACH

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Order Code and HW Options

Product Type:

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EXAMPLE: i101-1-256-E-PM-T-U

Consist Network	File System	FRAM	Service	Display Interface	Protocol	HW Int.	DSUB Bolt	
MVB i101	1 GB 8 GB	1 8 64 kB 256 kB 1 MB	64 256 1M	USB no <i><empty></i>	U no <i><empty></i>	DVI* no <i><empty></i>	V	
ETH i102*					Prozess Data Message Data Bus Administrator	P _ _ P M _ P _ B	EMD ESD+ T D	UNC M3 U M3
CAN i103*					TCP/UDP Sockets IPTCom Profinet RT EtherNet/IP – CPI	I B R C		
					J1939 CAN CANopen Master CANopenSlave	J L OM OS	120Ω 150Ω 12 15	UNC M3 U M3

Contact duagon for lead times and availability

Related Documents

Data Sheet i101	d-006273-nnnnnn.pdf	available at www.duagon.com
Data Sheet i102	d-006689-nnnnnn.pdf	
Data Sheet i103	d-006690-nnnnnn.pdf	

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